

Hierarchies of infinities and the ratio test



Outline

Hierarchies of infinities for sequences, part 1 and part 2

The ratio test theorem

Hierarchies of infinities for sequences, part 3 and part 4



Hierarchies of infinities for sequences, part 1 and part 2



Introduction: Hierarchies of infinities for sequences, part 1 and part 2

- ▶ Logarithms, powers and exponentials all diverge to $+\infty$, but at **different speeds**
- ▶ In this section we prove the first two parts of the **hierarchy of infinities**: the logarithm is negligible with respect to the power, the power with respect to the exponential
- ▶ These hierarchies quickly solve many indeterminate forms $\frac{\infty}{\infty}$

Theorem 1: Hierarchy of infinities (part I)

$$\lim_{n \rightarrow +\infty} \frac{\log_a n}{n^\alpha} = 0$$

for every $a > 1$ and $\alpha > 0$.

Proof (1/3)

We begin by establishing a useful inequality between any positive real number and its logarithm.

For $x \in \mathbb{R}$, $x > 0$, let k be the integer part of x , i.e., the integer k for which we have

$$k \leq x < k + 1, \quad \text{moreover we have } 2^x \geq 2^k = (1 + 1)^k \geq 1 + k > x,$$

Proof (2/3)

the first inequality follows from the monotonicity of the exponential function, the second from the binomial expansion (or from Bernoulli's inequality). Taking logarithms to base $a > 1$, we obtain

$$\log_a x < x \log_a 2 \quad (\text{for every } x \in \mathbb{R}, x > 0).$$

We now apply this inequality to the number $x = n^{\alpha/2}$, and we have

$$\frac{\alpha}{2} \log_a n < n^{\alpha/2} \log_a 2 \quad \text{and} \quad \frac{\log_a n}{n^{\alpha/2}} < \frac{2}{\alpha} \log_a 2,$$

Proof (3/3)

hence

$$\frac{\log_a n}{n^\alpha} = \frac{\log_a n}{n^{\alpha/2}} \frac{1}{n^{\alpha/2}} \leq \underbrace{\frac{2}{\alpha} \log_a 2}_{\text{constant}} \frac{1}{n^{\alpha/2}} \quad \text{and} \quad \frac{1}{n^{\alpha/2}} \rightarrow 0.$$

By the corollary of the comparison theorem, the claim follows. $\square$

Theorem 2: Hierarchy of infinities (part II)

$$\lim_{n \rightarrow +\infty} \frac{n^\alpha}{a^n} = 0$$

for every $a > 1$ and $\alpha > 0$.

Proof (1/3)

We use the hierarchy of infinities theorem (part I), replacing the integer n with the integer 2^n :

$$0 = \lim_{n \rightarrow +\infty} \frac{\log_a(2^n)}{(2^n)^\alpha} = \lim_{n \rightarrow +\infty} \frac{n \overbrace{\log_a 2}^{\text{constant}}}{(2^\alpha)^n} \quad \text{hence} \quad \frac{n}{(2^\alpha)^n} \rightarrow 0.$$

Proof (2/3)

If now $a > 1$ is fixed, choosing $\alpha > 0$ such that $2^\alpha = a$ we obtain that $\frac{n}{a^n} \rightarrow 0$, that is, the second relation in the special case where n is raised to the exponent 1. The general case follows from the identity:

$$\frac{n^\alpha}{a^n} = \left(\frac{n}{a^{n/\alpha}} \right)^\alpha = \left(\frac{n}{(a^{1/\alpha})^n} \right)^\alpha$$

Proof (3/3)

Indeed, by the previous result $\frac{n}{(a^{1/\alpha})^n} \rightarrow 0$ (the base $a^{1/\alpha}$ is still a number > 1), hence it follows that

$$\left(\frac{n}{(a^{1/\alpha})^n} \right)^\alpha \rightarrow 0 \text{ and from this the claim.}$$



- ▶ These limits describe the “speed” at which logarithms (with base > 1), powers (with exponent > 0) and exponentials (with base > 1) go to infinity. Logarithms with base > 1 go more slowly than any power with exponent > 0 ; powers with exponent > 0 go more slowly than any exponential with base > 1 .
- ▶ Exponentials with base > 1 are infinities of higher order than powers with exponent > 0 and than logarithms with base > 1 . Powers with exponent > 0 are infinities of higher order than logarithms with base > 1 .

Example: Computing limits using the hierarchy of infinities (1/2)

We compute the limit

$$\lim_{n \rightarrow +\infty} \sqrt[n]{n} = [\infty^0] \quad (\text{indeterminate form})$$

Now we write

$$\sqrt[n]{n} = n^{\frac{1}{n}} = e^{\log n^{1/n}} = e^{\frac{\log n}{n}}$$

Example: Computing limits using the hierarchy of infinities (2/2)

and studying the sequence in the exponent

$$a_n = \frac{\log n}{n} \quad \text{we have} \quad a_n \rightarrow 0$$

thanks to the hierarchy of infinities theorem. Hence we have:

$$\lim_{n \rightarrow +\infty} \sqrt[n]{n} = \lim_{n \rightarrow +\infty} e^{\frac{\log n}{n}} = 1$$

Example: Computing limits using the hierarchy of infinities (1/3)

We compute the limit

$$\lim_{n \rightarrow +\infty} \frac{2^n + n}{2^{n+1}} = \left[\frac{\infty}{\infty} \right] \quad (\text{indeterminate form})$$

We can write:

$$\underbrace{2^n + n}_{a_n} = \underbrace{2^n}_{a'_n} \underbrace{\left(1 + \frac{n}{2^n}\right)}_{\rightarrow 1} \sim 2^n$$

Example: Computing limits using the hierarchy of infinities (2/3)

since, thanks to the hierarchy of infinities theorem, we have

$$\frac{n}{2^n} \rightarrow 0,$$

i.e., 2^n is an infinity of higher order than n . Hence we can write:

$$\frac{2^n + n}{2^{n+1}} \sim \frac{2^n}{2^{n+1}}$$

Example: Computing limits using the hierarchy of infinities (3/3)

Now, since

$$\frac{2^n}{2^{n+1}} = \frac{2^n}{2^n 2} = \frac{1}{2}$$

then

$$\lim_{n \rightarrow +\infty} \frac{2^n + n}{2^{n+1}} = \frac{1}{2}$$

Summary

Hierarchy of infinities (parts I and II):

- ▶ $\lim_{n \rightarrow +\infty} \frac{\log_a n}{n^\alpha} = 0$ for every $a > 1$, $\alpha > 0$
- ▶ $\lim_{n \rightarrow +\infty} \frac{n^\alpha}{a^n} = 0$ for every $a > 1$, $\alpha > 0$
- ▶ When computing limits we factor out the infinity of highest order and neglect the others

The ratio test theorem



Introduction: The ratio test theorem

- ▶ In this section we introduce the **ratio test**: for a positive sequence, the limit of $\frac{a_{n+1}}{a_n}$ tells whether a_n tends to 0 or to $+\infty$
- ▶ It is particularly useful for sequences with factorials and exponentials
- ▶ We will use it to complete the hierarchy of infinities

Theorem 3: Ratio test

Let $\{a_n\}$ be a positive sequence (i.e., $a_n > 0$ for every n).

If there exists $\lim_{n \rightarrow +\infty} \frac{a_{n+1}}{a_n} = l$ and $\begin{cases} l < 1, & \text{then } a_n \rightarrow 0 \\ l > 1 \text{ (or } l = +\infty), & \text{then } a_n \rightarrow +\infty \end{cases}$

The ratio test theorem

- ▶ The previous theorem reduces the study of the limit of a positive sequence $\{a_n\}$ to the computation of the limit of another sequence, the sequence (of ratios)

$$n \mapsto \frac{a_{n+1}}{a_n}$$

In some cases the latter is easier to study than the original one, as we will see in the examples.

Note that in the case $l = 1$ the theorem does not allow us to conclude anything.

Proof (1/5)

1. Suppose that:

$$\frac{a_{n+1}}{a_n} \rightarrow l < 1$$

Then, for every $\varepsilon > 0$, we have, for $n \geq n(\varepsilon)$,

$$\frac{a_{n+1}}{a_n} < l + \varepsilon$$

Proof (2/5)

We can then write the chain of inequalities:

$$a_{n(\varepsilon)+1} < (1 + \varepsilon) a_{n(\varepsilon)}, \quad a_{n(\varepsilon)+2} < (1 + \varepsilon) \underbrace{a_{n(\varepsilon)+1}}_{< (1+\varepsilon) a_{n(\varepsilon)}} < (1 + \varepsilon)^2 a_{n(\varepsilon)}, \quad \dots$$

hence

$$a_{n(\varepsilon)+k} < (1 + \varepsilon)^k a_{n(\varepsilon)}$$

Choosing ε small enough to have $1 + \varepsilon < 1$, we have

$$(1 + \varepsilon)^k \rightarrow 0 \quad \text{for } k \rightarrow +\infty$$

Proof (3/5)

On the other hand, $n(\varepsilon)$ is fixed and consequently $a_{n(\varepsilon)}$ is fixed as well; hence for k large enough the right-hand side (and therefore the left-hand side) is as small as we like. This proves the first claim, that is: $a_n \rightarrow 0$.

2. Suppose that:

$$\frac{a_{n+1}}{a_n} \rightarrow l > 1$$

Proof (4/5)

Then, for every $\varepsilon > 0$, we have, for $n \geq n(\varepsilon)$,

$$\frac{a_{n+1}}{a_n} > 1 - \varepsilon$$

We choose ε small enough to have $1 - \varepsilon > 1$; with steps similar to before we can write

$$a_{n(\varepsilon)+k} > (1 - \varepsilon)^k a_{n(\varepsilon)} \quad \text{and} \quad (1 - \varepsilon)^k \rightarrow +\infty \quad \text{for} \quad k \rightarrow +\infty.$$

Proof (5/5)

On the other hand, $n(\varepsilon)$ is fixed and consequently $a_{n(\varepsilon)}$ is fixed as well; hence for k large enough the right-hand side (and therefore the left-hand side) is as large as we like. This proves the second claim, that is: $a_n \rightarrow +\infty$.



Example: Using the ratio test theorem (1/3)

Let us try to compute, with the ratio test, the limit

$$\lim_{n \rightarrow +\infty} \frac{\log n}{n} = \left[\frac{\infty}{\infty} \right] \quad (\text{indeterminate form})$$

we have

$$\frac{a_{n+1}}{a_n} = \frac{\log(n+1)}{n+1} \cdot \frac{n}{\log n} = \frac{n}{n+1} \cdot \frac{\log(n+1)}{\log n}$$

Example: Using the ratio test theorem (2/3)

we have

$$\frac{n}{n+1} = \frac{n+1-1}{n+1} = 1 - \underbrace{\frac{1}{n+1}}_{\rightarrow 0} \quad \text{hence} \quad \lim_{n \rightarrow +\infty} \frac{n}{n+1} = 1$$

moreover we have

$$\frac{\log(n+1)}{\log n} \sim \frac{\log n}{\log n} = 1 \quad \text{hence} \quad \lim_{n \rightarrow +\infty} \frac{\log(n+1)}{\log n} = 1$$

where $\log(n+1) \sim \log(n)$ by the substitution principle. Now, using the theorem on the algebra of limits, we have

$$\lim_{n \rightarrow +\infty} \frac{a_{n+1}}{a_n} = 1 \cdot 1 = 1$$

Example: Using the ratio test theorem (3/3)

hence Theorem (ratio test) does not allow us, in this case, to conclude anything.



Summary

Ratio test:

- ▶ Let $a_n > 0$ and $\frac{a_{n+1}}{a_n} \rightarrow l$
- ▶ If $l < 1$ then $a_n \rightarrow 0$
- ▶ If $l > 1$ (or $l = +\infty$) then $a_n \rightarrow +\infty$
- ▶ If $l = 1$ the test is inconclusive

Hierarchies of infinities for sequences, part 3 and part 4



Introduction: Hierarchies of infinities for sequences, part 3 and part 4

- ▶ In this section we complete the hierarchy of infinities with the **factorial** $n!$ and with n^n
- ▶ The proofs are direct applications of the ratio test

Theorem 4: Hierarchy of infinities (part III)

$$\lim_{n \rightarrow +\infty} \frac{a^n}{n!} = 0$$

for every $a > 0$.

- ▶ hence exponentials with base > 0 go more slowly than the factorial

Proof

We apply the ratio test to the sequence

$$b_n = \frac{a^n}{n!} \quad \text{we have} \quad \frac{b_{n+1}}{b_n} = \frac{a^{n+1}}{(n+1)!} \cdot \frac{n!}{a^n} = \frac{a}{n+1} \quad \text{and} \quad \frac{a}{n+1} \rightarrow 0$$

Using the ratio test theorem, we obtain the claim. □

Theorem 5: Hierarchy of infinities (part IV)

$$\lim_{n \rightarrow +\infty} \frac{n!}{n^n} = 0$$

- ▶ hence the factorial goes more slowly than n^n

Proof (1/2)

We apply the ratio test to the sequence

$$b_n = \frac{n!}{n^n}$$

we have

$$\begin{aligned}\frac{b_{n+1}}{b_n} &= \frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!} = \frac{(n+1) \cdot n!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!} \\ &= \frac{(n+1) \cdot n^n}{(n+1)^n \cdot (n+1)} = \left(\frac{n}{n+1} \right)^n = \frac{1}{\left(1 + \frac{1}{n}\right)^n}\end{aligned}$$

Proof (2/2)

and since

$$\left(1 + \frac{1}{n}\right)^n \rightarrow e$$

then

$$\frac{b_{n+1}}{b_n} \rightarrow \frac{1}{e} < 1$$

Using the ratio test theorem, we obtain the claim. □

Summary

Hierarchy of infinities (parts III and IV):

▶ $\lim_{n \rightarrow +\infty} \frac{a^n}{n!} = 0$ for every $a > 0$

▶ $\lim_{n \rightarrow +\infty} \frac{n!}{n^n} = 0$

Summary

Complete hierarchy of infinities ($a > 1$, $\alpha > 0$):

$$\log_a n \ll n^\alpha \ll a^n \ll n! \ll n^n$$

where $x_n \ll y_n$ means $\frac{x_n}{y_n} \rightarrow 0$

- ▶ Parts I and II are proved directly, parts III and IV with the **ratio test**
- ▶ Ratio test: if $\frac{a_{n+1}}{a_n} \rightarrow l < 1$ then $a_n \rightarrow 0$; if $l > 1$ then $a_n \rightarrow +\infty$